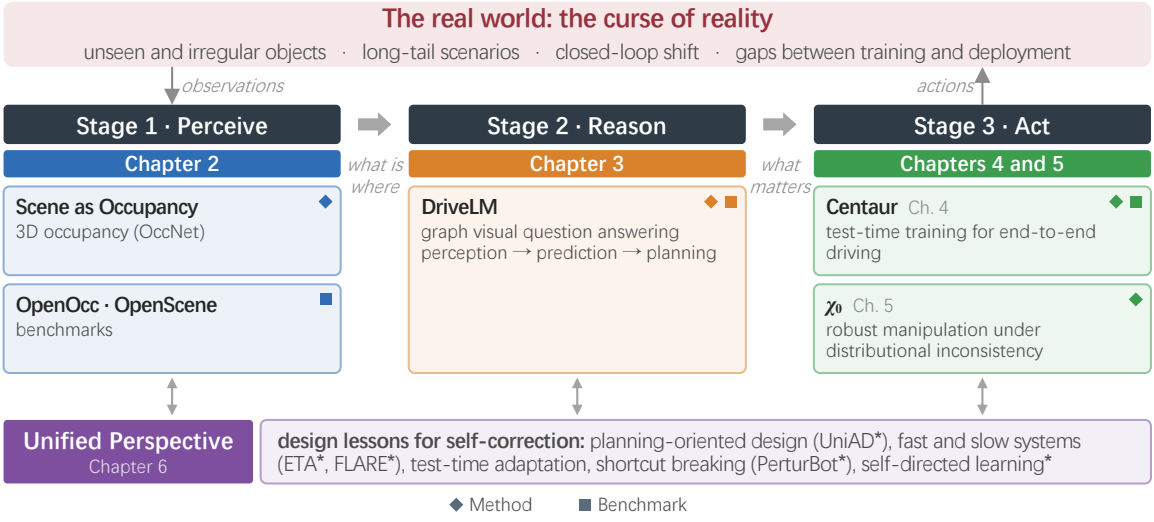

The real world: the curse of reality
unseen and irregular objects · long-tail scenarios · closed-loop shift · gaps between training and deployment
observations
actions
Stage 1 · Perceive
Stage 2 · Reason
Stage 3 · Act
Chapter 2
what is where
Chapter 3
what matters
Chapters 4 and 5
Scene as Occupancy
3D occupancy (OccNet)
OpenOcc · OpenScene
benchmarks
DriveLM
graph visual question answering
perception → prediction → planning
Centaur Ch. 4
test-time training for end-to-end driving
χ0 Ch. 5
robust manipulation under distributional inconsistency
Unified Perspective
Chapter 6
design lessons for self-correction: planning-oriented design (UniAD*), fast and slow systems (ETA*, FLARE*), test-time adaptation, shortcut breaking (PerturBot*), self-directed learning*
Method
Benchmark